HUMUS HERO – THE WATER SAVING SOIL

MOTTO: “More Humus, More Water, Healthier Plants.”

COMPONENTS: - The project uses an Arduino uno, soil moisture sensor, buzzer, 5v battery, connecting wires, dry soil, and soil mixed with humus or compost.

To construct the project, at first code is uploaded to Arduino uno and then, soil moisture sensor and buzzer is connected to Arduino uno, to power up Arduino uno 5v battery is used. Two containers are prepared: one with dry soil and another with soil mixed with humus. The sensor is inserted into each container to check the moisture level.

When the sensor is placed in dry soil, the moisture level is low the Arduino uno makes the buzzer beeps, indicating that the soil needs water. When the sensor is placed in humus-rich soil, the soil retains more water, so the moisture level is higher and the buzzer does not beep.

This project is used to compare the water-holding capacity of different types of soil. It demonstrates that humus improves soil quality and helps maintain moisture for plant growth.

CONCLUSION: Humus Hero shows that adding humus or compost to soil can reduce water loss and keep plants healthy for a longer time. This method can help farmers save water, improve soil fertility, and reduce frequent irrigation. It is a low-cost and eco-friendly.